

June 17, 2026

July 17, 2026

University of Southampton, Southampton, United Kingdom.

Dear Admissions Committee,

I am writing to express my strong interest in the PhD position in AI and Sustainable Technologies at the University of Southampton, focusing on energy-efficient computing using nanostructured materials. As a data science engineer with a passion for innovative solutions, I am confident that this project aligns perfectly with my skills and research interests.

Throughout my undergraduate studies, I have consistently demonstrated my ability to excel in multidisciplinary projects that integrate AI, machine learning, and computer vision. One of my most notable achievements is the development of *MedicalMambaTS*, a linear – time state space model for sub – millisecond bedside inference. This achievement has direct implications for energy – efficient computing, as it enables real – time decision – making in resource – constrained environments.

I am particularly drawn to this project because of its emphasis on designing and fabricating nanoscale neural elements using emerging semiconductors. My experience with PyTorch Autograd has given me a strong foundation in programming language design and its application to complex problems. I believe that my expertise in developing physics-informed models, as showcased by *PI_DeepONet*, would be highly valuable in this project.

I am excited about the prospect of working within the SustAI CDT's multidisciplinary environment and collaborating with researchers across various disciplines. My experience with cloud forecasting using 100Hz VitalDB telemetry has taught me the importance of robust data pipelines and efficient processing. I am confident that my skills and knowledge would be a valuable addition to your team.

Thank you for considering my application. I look forward to discussing this opportunity further.

Sincerely,

Ismail Aissaoui